



# Exercise 1

## Variables, Collections & Conditionals

Data Analytics with Python · Databricks Environment

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### Instructions

Work through each section in your Databricks notebook. Create a new cell for each task. Read each question carefully before writing your code.

- You may refer to the notes and examples shown in class.
- Use `print()` to display your results where asked.
- Write your answers (expected output) in the spaces provided below.
- Complete this document using Microsoft word or google documents.
- Save the document as PDF
- Upload both the PDF and databricks notebook to GitHub
- Send GitHub link, name, course name to (mvuko@brightlearn.co.za)

### The Dataset

Below is a small student dataset. You will manually enter this data as Python variables and collections in your notebook. Each row represents one student's record.

| student_name | age | subject   | score | passed |
|--------------|-----|-----------|-------|--------|
| "Amara"      | 19  | "Math"    | 72    | True   |
| "Ben"        | 21  | "Science" | 55    | False  |
| "Chloe"      | 20  | "Math"    | 88    | True   |
| "David"      | 22  | "English" | 45    | False  |
| "Eva"        | 19  | "Science" | 91    | True   |
| "Farai"      | 23  | "English" | 63    | True   |
| "Grace"      | 20  | "Math"    | 50    | False  |
| "Hassan"     | 21  | "Science" | 77    | True   |

**Note:** This is the only data you will work with in this exercise.



## Section 1 — Variables & Data Types

In this section, you will create individual variables for a single student's record and explore Python data types.

### Task 1.1 — Create variables for Amara's record

In a new cell, create the following variables exactly as shown:

```
student_name = "Amara"  
age          = 19  
subject      = "Math"  
score        = 72  
passed       = True
```

**Task 1.1** After running the cell, what is the data type of each variable? Write it next to each variable name.

- a. student\_name → \_\_\_\_ **str** \_\_\_\_
- b. age → \_\_\_\_ **int** \_\_\_\_
- c. score → \_\_\_\_ **int** \_\_\_\_
- d. passed → \_\_\_\_ **bool** \_\_\_\_

Your answer:

### Task 1.2 — Use type() and print()

Add a new cell and use the type() function to confirm the data type of score and passed. Then use print() to display the result.

**Task 1.2** Write the code you used and record the output below.

- e. Code: `_print(type(score))` \_\_\_\_ `_print(type(passed))` \_\_\_\_
- f. Output for type(score): \_\_\_\_ **int** \_\_\_\_
- g. Output for type(passed): \_\_\_\_ **bool** \_\_\_\_

Your answer:



### Task 1.3 — Simple arithmetic

A student's final mark is calculated as their score divided by 100, multiplied by 150 (the exam is out of 150).

**Task 1.3** Write a single line of Python that calculates Amara's final mark and stores it in a variable called `final_mark`. Then print it.

- h. What is Amara's `final_mark`? 108.0
- i. What data type is `final_mark`? Float

Your answer:

## Section 2 — Collections

In this section you will store the dataset in Python lists and a dictionary, and access specific values.

### Task 2.1 — Create a list of student names

Create a list called `student_names` that contains all 8 student names from the dataset, in order.

**Task 2.1** Answer the following questions about your list:

- j. What is `student_names[0]`? Amara
- k. What is `student_names[-1]`? Hassan
- l. What is `student_names[2:5]`? ['Chloe', 'David', 'Eva']
- m. What is `len(student_names)`? 8

Your answer:

### Task 2.2 — Create a list of scores

Create a list called `scores` containing all 8 scores in the same order as the names list.



**Task 2.2** Using your scores list, answer:

- n. What is scores[0]? \_\_ **72** \_\_\_\_\_
- o. What is scores[4]? \_\_ **91** \_\_\_\_\_
- p. What is the result of scores[0] + scores[4]? \_\_ **163** \_\_\_\_\_

Your answer:

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### Task 2.3 — Create a dictionary for one student

Create a dictionary called hassan that stores all five fields for Hassan as key-value pairs.

```
hassan = {  
    "student_name": ...,  
    "age": ...,  
    "subject": ...,  
    "score": ...,  
    "passed": ...  
}
```

**Task 2.3** Using your dictionary, answer:

- q. What does hassan["subject"] return? \_\_ **Science** \_\_\_\_\_
- r. What does hassan["score"] + 10 return? \_\_ **87** \_\_\_\_\_
- s. What does type(hassan["passed"]) return? \_\_ **True** \_\_\_\_\_

Your answer:

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### Task 2.4 — A list of dictionaries

Create a list called students where each element is a dictionary representing one student. Include only Amara, Ben, Chloe, and David.



**Task 2.4** Using your list of dictionaries, answer:

- t. What is students[1]["score"]? \_\_ **Ben** \_\_
- u. What is students[0]["student\_name"]? \_\_ **Amara** \_\_
- v. How many items are in students? \_\_ **4** \_\_

Your answer:

## Section 3 — Logical Operators & Conditionals

In this section you will use comparison operators, logical operators (and, or, not), and if/elif/else statements to analyse student records.

### Task 3.1 — Comparison expressions

Using the variables you created in Section 1 (Amara's record), evaluate the following expressions. Write the result (True or False) without running the code first — then verify in your notebook.

**Task 3.1** Predict the output of each expression:

- w. score > 60 → \_\_ **True** \_\_
- x. age == 20 → \_\_ **False** \_\_
- y. subject != "Science" → \_\_ **True** \_\_
- z. score >= 72 → \_\_ **True** \_\_

Your answer:

### Task 3.2 — Logical operators

Still using Amara's variables, evaluate these combined expressions:



**Task 3.2** Predict and verify:

- aa. score > 60 and passed == True → True
- bb. age < 18 or score > 70 → True
- cc. not passed → False
- dd. score > 80 and subject == "Math" → False

Your answer:

**Task 3.3 — Write an if/elif/else statement**

Write a conditional statement that prints a grade category based on a student's score. Use the following rules:

| Score Range  | Grade       |
|--------------|-------------|
| 75 and above | Distinction |
| 50 – 74      | Pass        |
| Below 50     | Fail        |

Test your conditional using score = 72 (Amara) and again with score = 45 (David).

**Task 3.3** Write your if/elif/else code below and record the output for each test:

- ee. Output when score = 72: Pass
- ff. Output when score = 45: Fail
- gg. Output when score = 91: Distinction

Your answer:

**Task 3.4 — Challenge: combined condition**

Write a conditional statement that checks two conditions at once: the student scored above 60 AND their subject is "Math". If both are true, print "Math high achiever". Otherwise print "Does not meet criteria".



**Task 3.4** Test with Amara (score=72, subject="Math") and Ben (score=55, subject="Science"):

- hh. Output for Amara: Math high achieve
- ii. Output for Ben: Does not meet criteria
- jj. What logical operator did you use? AND

Your answer:

## Reflection

Answer these short questions in your own words — no code needed.

**Task R1** What is the difference between a list and a dictionary in Python? Give one example of when you would choose each.

Your answer:

**A list stores items in a specific order. Use a list when you have a collection of similar items like Storing test scores.**

**A dictionary stores data as key-value pairs. Use a dictionary when you want to label data like Student records.**

**Task R2** In Task 3.2, explain in plain English what this expression checks: `score > 60 and passed == True`

Your answer:

**checks two conditions at the same time:**

**if the score is greater than 60 and if passed is True**

**Both conditions must be true for the whole expression to be true.**



**Task R3** If you wanted to store all 8 students and be able to look up any student by name quickly, which collection type would you use? Why?

Your answer:

**I would use a dictionary because it allows you to quickly look up a student using their name as the key.**

*Good luck!*